Rearranging Variables & Logarithms

Inverse Operations, Exponentials, and Casio Workflows

M. Sc. Guillermo Gachuz Atitlán

October 20, 2026

Churchill College | IGCSE

MSc Gachuz Atitlán

Session Flow: Rearranging Variables & Logarithms

A 100-minute student-centered sequence based on Buckingham's PGC Framework.

Lesson Roadmap (100 min)

- **10 min – Phase 1:** Novelty & Curiosity (The "Inverse Machine" Puzzle)
- **20 min – Phase 2:** Expectation & Motivation (SADMEP Steps, Log Anatomy & Casio Guide)
- **45 min – Phase 3:** Effort & Scaffolding (Level 1, Level 2 & Level 3 Challenges)
- **25 min – Phase 4:** Recognition & Peer Feedback (Notebook Swap & Synthesis)

The Black Box Challenge

"Every mathematical operation has an 'antidote' (inverse operation) that undoes it. Can you identify all the pairs?"

Match the Antidote

- Addition $(+a) \leftrightarrow$

The Black Box Challenge

"Every mathematical operation has an 'antidote' (inverse operation) that undoes it. Can you identify all the pairs?"

Match the Antidote

- Addition $(+a)$ $\leftrightarrow$ Subtraction $(-a)$
- Multiplication $(\times b)$ $\leftrightarrow$

The Black Box Challenge

"Every mathematical operation has an 'antidote' (inverse operation) that undoes it. Can you identify all the pairs?"

Match the Antidote

- Addition $(+a) \leftrightarrow$ Subtraction $(-a)$
- Multiplication $(\times b) \leftrightarrow$ Division $(\div b)$
- Power $(x^n) \leftrightarrow$

The Black Box Challenge

"Every mathematical operation has an 'antidote' (inverse operation) that undoes it. Can you identify all the pairs?"

Match the Antidote

- Addition $(+a)$ $\leftrightarrow$ Subtraction $(-a)$
- Multiplication $(\times b)$ $\leftrightarrow$ Division $(\div b)$
- Power (x^n) $\leftrightarrow$ n-th Root $(\sqrt[n]{x})$

The Black Box Challenge

"Every mathematical operation has an 'antidote' (inverse operation) that undoes it. Can you identify all the pairs?"

Match the Antidote

- Addition $(+a)$ $\leftrightarrow$ Subtraction $(-a)$
- Multiplication $(\times b)$ $\leftrightarrow$ Division $(\div b)$
- Power (x^n) $\leftrightarrow$ n-th Root $(\sqrt[n]{x})$

The Unsolved Mystery

How do we undo a variable in the exponent?

$$2^x = 4 \quad \Longrightarrow \quad x = \,?$$

Discuss with your partner: What operation "lowers" x?

2. Expectation & Motivation | 4 Steps to Rearrange Variables

To isolate any variable x, follow the **SADMEP** rule (PEMDAS in reverse order):

1. **Step 1 – Identify:** Locate x and identify all operations affecting it.
2. **Step 2 – Rank:** List operations from innermost (closest to x) to outermost (furthest from x).
3. **Step 3 – Undo (SADMEP):** Remove operations in reverse order using their **inverse operations**.
4. **Step 4 – Simplify:** Compute and simplify the resulting expression.

Always peel off operations from the outside in!

2. Guided Practice: Rearranging Variables

Isolate the target variable x in each case using the **SADMEP** strategy.

Level 1: Easy

(1–2 Linear Steps)

1. $4x - 9 = 15$
2. $y = 3x + b$
3. $\frac{x}{6} + 4 = 9$

Level 2: Medium

(Powers & Roots)

1. $2x^2 - 5 = 45$
2. $\sqrt{3x+1} = 5$
3. $A = \pi x^2 + c$

Level 3: Hard

(Algebraic Fractions & x on both sides)

1. $v^2 - x = u^2 + 2ax$
2. $y = \frac{2x+1}{x-3}$
3. $T = 2\pi\sqrt{\frac{x}{g}}$

Success Criteria

Apply inverse operations in reverse order (SADMEP). Show all intermediate steps in your notebook!

Asking *"What power of 2 gives 4?"* ($2^x = 4$) is identical to writing a **logarithm**:

Color-Coded Equivalence

$$b^x = a \iff \log_b(a) = x$$

How to read this out loud:

"To what power must I raise the base (b) to obtain the value inside parentheses (a)? The answer is x."

Asking *"What power of 2 gives 4?"* ($2^x = 4$) is identical to writing a **logarithm**:

Color-Coded Equivalence

$$b^x = a \iff \log_b(a) = x$$

How to read this out loud:

"To what power must I raise the base (b) to obtain the value inside parentheses (a)? The answer is x."

Quick Verbal Intuition Examples:

$$\log_{10}(1000) = 3$$

$10^3 = 1000$

$$\log_2\left(\frac{1}{4}\right) = -2$$

$2^{-2} = \frac{1}{4}$

$$\log_\pi(1) = 0$$

$\pi^0 = 1$

Euler's Number (e)

- $e \approx 2.71828...$ is a fundamental mathematical constant (like π).
- It governs continuous growth and decay models.

Natural Logarithm (ln)

- $\ln(x)$ stands for **Natural Logarithm**.
- It is simply a logarithm with base e:

$$\ln(x) \equiv \log_e(x)$$

Euler's Number (e)

- $e \approx 2.71828...$ is a fundamental mathematical constant (like π).
- It governs continuous growth and decay models.

Logarithm Base 10 (log)

- $\log(x)$ stands for **Logarithm Base 10**.
- It is simply a logarithm with base 10:

$$\log(x) \equiv \log_{10}(x)$$

Natural Logarithm (ln)

- $\ln(x)$ stands for **Natural Logarithm**.
- It is simply a logarithm with base e:

$$\ln(x) \equiv \log_e(x)$$

Inverse Relationship

$$e^x = y \iff x = \ln(y)$$

$$b^x = y \iff x = \log_b(y)$$

Apply inverse operations to isolate x: **Exponentials and Logarithms undo each other!**

Example A: Isolating the Exponent

Solve for x:

$$3e^{2x} = 12$$

1. Divide by 3:

$$e^{2x} = 4$$

Apply inverse operations to isolate x: **Exponentials and Logarithms undo each other!**

Example A: Isolating the Exponent

Solve for x:

$$3e^{2x} = 12$$

1. Divide by 3:

$$e^{2x} = 4$$

2. Apply ln and divide by 2:

$$2x = \ln(4) \implies x = \frac{\ln(4)}{2}$$

Apply inverse operations to isolate x: **Exponentials and Logarithms undo each other!**

Example A: Isolating the Exponent

Solve for x:

$$3e^{2x} = 12$$

1. Divide by 3:

$$e^{2x} = 4$$

2. Apply ln and divide by 2:

$$2x = \ln(4) \implies x = \frac{\ln(4)}{2}$$

Example B: Isolating the Logarithm

Solve for x:

$$\ln(x-1) + 2 = 5$$

1. Subtract 2:

$$\ln(x-1) = 3$$

Apply inverse operations to isolate x: **Exponentials and Logarithms undo each other!**

Example A: Isolating the Exponent

Solve for x:

$$3e^{2x} = 12$$

1. Divide by 3:

$$e^{2x} = 4$$

2. Apply ln and divide by 2:

$$2x = \ln(4) \implies x = \frac{\ln(4)}{2}$$

Example B: Isolating the Logarithm

Solve for x:

$$\ln(x-1) + 2 = 5$$

1. Subtract 2:

$$\ln(x-1) = 3$$

2. Exponentiate with base e and add 1:

$$x - 1 = e^3 \implies x = e^3 + 1$$

Step-by-Step Casio Menu Navigation

1. Press **MENU** and select RUN-MAT using **EXE**.
2. Press the **MATH: F4** menu.
3. Now select **logab: F2** to access the logarithm function.
4. Fill in the required values, base and argument: $\log_{\square}(\square)$.

Casio Verification Practice

Try calculating $\log_2(32)$ on your calculator:

Step-by-Step Casio Menu Navigation

1. Press **MENU** and select RUN-MAT using **EXE**.
2. Press the **MATH: F4** menu.
3. Now select **logab: F2** to access the logarithm function.
4. Fill in the required values, base and argument: $\log_{\square}(\square)$.

Casio Verification Practice

Try calculating $\log_2(32)$ on your calculator: **Result** $= 5$ (since $2^5 = 32$).

[Level 1: Easy – Pair Work | 10 min]

Goal: Direct conversions, basic variable isolation, and calculator verification.

Task 1: Convert from Exponential to Logarithmic form (and vice versa):

1. $3^4 = 81 \implies$ **?**
2. $10^3 = 1000 \implies$ **?**

1. $\log_2(64) = 6 \implies$ **?**
2. $\ln(e^5) = 5 \implies$ **?**

Task 2: Solve directly using Casio:

a) $\log_5(125)$ b) $\log_{10}(0.01)$ c) $\ln(20)$

[Level 2: Medium – Think-Pair-Share | 15 min]

Goal: Master operational order when undoing expressions.

Spot the Mistake!

A student attempted to solve $10^{x+1} = 4$ as follows:

$$\text{Step 1: } 10^x = 4 - 1 = 3 \quad \Longrightarrow \quad \text{Step 2: } x = \log_{10}(3)$$

Question: Why is Step 1 incorrect? What is the outermost operation?

[Level 2: Medium – Think-Pair-Share | 15 min]

Goal: Master operational order when undoing expressions.

Spot the Mistake!

A student attempted to solve $10^{x+1} = 4$ as follows:

$$\text{Step 1: } 10^x = 4 - 1 = 3 \quad \Longrightarrow \quad \text{Step 2: } x = \log_{10}(3)$$

Question: Why is Step 1 incorrect? What is the outermost operation?

Correct Modelled Steps

Solve: $10^{x+1} = 4$

- Undo base 10: $x + 1 = \log_{10}(4)$
- Undo $+1$: $x = \log_{10}(4) - 1$

[Level 3: Hard – Team Challenge | 20 min]

Goal: Apply SADMEP to multi-step exponential and logarithmic equations.

Solve for x step-by-step in your notebook:

Exponential Equations:

1. $3^{x^2} = 3$
2. $5^{2x-2} = 1$
3. $4e^{3x-1} = 20$

Logarithmic Equations:

1. $\log_{10}(x^2) = 10$
2. $\log_5(2x + 2) = 20$
3. $\ln(x^2 - 3) = 0$

4. Recognition & Feedback | Peer Verification Protocol

Notebook Exchange (10 min): Swap notebooks with a neighboring pair.

Verification Checklist for Level 3 Solutions

- ✓ **Step 1 & 2:** Did they correctly list/identify the outermost operation?
- ✓ **Step 3:** Did they apply the inverse operation correctly (e.g., exponentiation vs. logarithm)?
- ✓ **Step 4:** Is the algebraic simplification accurate?

1. **Specific Praise:** Highlight a clean, well-structured step.
2. **Constructive Question:** Point out any order-of-operation oversight.
3. **Achievement Signature:** Sign at the bottom of their page upon verification.

Session Synthesis

Operation	Inverse Operation	Example Isolation
Addition $(+a)$	Subtraction $(-a)$	$x+3=7 \implies x=7-3$
Multiplication $(\times b)$	Division $(\div b)$	$5x=20 \implies x=\frac{20}{5}$
Exponentiation (b^x)	Logarithm $(\log_b)$	$10^x=4 \implies x=\log_{10}(4)$
Logarithm $(\log_b(x))$	Base Exponent $(b^{(\cdot)})$	$\log_2(x)=3 \implies x=2^3$
Natural Base (e^x)	Natural Log (ln)	$e^x=5 \implies x=\ln(5)$

Great job mastering variable rearrangements today!

Advanced Rearranging & Factoring

Algebraic Fractions, Logarithm Properties, and Factoring Strategies

M. Sc. Guillermo Gachuz Atitlán

October 20, 2026

Churchill College | IB Mathematics & IGCSE

MSc Gachuz Atitlán

Session Flow: Advanced Rearranging (50 min)

Mastering the 3 most challenging scenarios when isolating x.

Lesson Agenda (50 min)

- **05 min – Intro:** The 3 hard cases in algebraic rearrangements.
- **12 min – Case 1:** Algebraic fractions with variables in top & bottom.
- **15 min – Case 2:** Logarithm properties & exponential equations ($a^x = b^x$).
- **13 min – Case 3:** Factoring toolset & quadratic-form equations.
- **05 min – Wrap-Up:** Practice recap & synthesis.

Case 1: Variable x in Both Numerator and Denominator

The Challenge

Isolate x when it appears in both the top and bottom of a rational function:

$$y = \frac{ax + b}{cx + d}$$

Case 1: Variable x in Both Numerator and Denominator

The Challenge

Isolate x when it appears in both the top and bottom of a rational function:

$$y = \frac{ax + b}{cx + d}$$

4-Step Strategy

1. **Clear the fraction:** Multiply both sides by the denominator $(cx + d)$.
2. **Expand:** Distribute to remove parentheses.
3. **Collect:** Move all terms containing x to one side and non-x terms to the other.
4. **Factor out & Divide:** Factor out x as a common term, then isolate it.

Case 1: Worked Example

Example: Rearrange $y = \frac{2x+1}{x-3}$ to make x the subject.

Step-by-Step

1. Multiply by $(x-3)$:
$$y(x-3) = 2x+1$$
2. Expand:
$$yx - 3y = 2x + 1$$
3. Move x-terms to the left:
$$yx - 2x = 3y + 1$$

Final Steps

1. Factor out x:
$$x(y-2) = 3y+1$$
2. Divide by $(y-2)$:
$$x = \frac{3y+1}{y-2}$$

Notice this process also gives us the inverse function $f^{-1}(y)$!

Case 2: Essential Logarithm Laws

Before solving equations with variables in exponents, let's review key logarithm rules:

Case 2: Essential Logarithm Laws

Before solving equations with variables in exponents, let's review key logarithm rules:

Fundamental Logarithm Laws

Power Law	$\log_b(A^k) = k \cdot \log_b(A)$ (Brings x down!)
Product Law	$\log_b(A \cdot B) = \log_b(A) + \log_b(B)$
Quotient Law	$\log_b\left(\frac{A}{B}\right) = \log_b(A) - \log_b(B)$
Base Identities	$\ln(e) = 1, \quad \log_{10}(10) = 1, \quad \log_b(b) = 1$

Case 2: Essential Logarithm Laws

Before solving equations with variables in exponents, let's review key logarithm rules:

Fundamental Logarithm Laws

Power Law	$\log_b(A^k) = k \cdot \log_b(A)$ (Brings x down!)
Product Law	$\log_b(A \cdot B) = \log_b(A) + \log_b(B)$
Quotient Law	$\log_b\left(\frac{A}{B}\right) = \log_b(A) - \log_b(B)$
Base Identities	$\ln(e) = 1, \quad \log_{10}(10) = 1, \quad \log_b(b) = 1$

The Goal

Isolate x when it appears in the exponents of different bases (e.g., $a^x = b^{x+1}$).

Case 2: Step-by-Step Strategy for $a^{mx+d} = b^{nx+c}$

Universal 3-Step Recipe

1. **Apply Logarithms:** Apply a logarithm (ln or $\log_b$) to get rid of one of the exponential terms and the apply the properties:
   - $\log_b(A \cdot B) = \log_b A + \log_b B$ (logarithm of a product)
   - $\log_b\left(\frac{A}{B}\right) = \log_b A - \log_b B$ (logarithm of a quotient)
   - $\log_b(A^k) = k \cdot \log_b A$ (logarithm of a power)

   to bring down the variable exponents.
2. **Expand & Group:** Expand any algebraic products and group all terms containing x on one side of the equation.
3. **Factor Out & Isolate:** Factor out x as a common factor and divide both sides by the remaining algebraic expression to solve for x.

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

3. Expand:

There is nothing to expand

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

3. Expand:

There is nothing to expand

Final Steps

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

3. Expand:

There is nothing to expand

Final Steps

1. Collect x-terms:

$$x - x\log_2(5) = -1$$

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

3. Expand:

There is nothing to expand

Final Steps

1. Collect x-terms:

$$x - x\log_2(5) = -1$$

2. Factor out x:

$$x(1 - \log_2(5)) = -1$$

Case 2: Equations of Form $a^x = b^x$ or $a^{mx} = b^{nx+c}$

Example: Solve for x in the equation $2^{x+1} = 5^x$

Step-by-Step

1. Apply $\log_2$:

$$x + 1 = \log_2(5^x)$$

2. Apply Power Law:

$$x + 1 = x\log_2(5)$$

3. Expand:

There is nothing to expand

Final Steps

1. Collect x-terms:

$$x - x\log_2(5) = -1$$

2. Factor out x:

$$x(1 - \log_2(5)) = -1$$

3. Isolate x:

$$x = \frac{-1}{1 - \log_2(5)}$$

Case 3: Factoring Toolset for Rearrangements

Factoring breaks expressions into products, making it possible to isolate x or set factors to zero.

1. Common Factor

Extract identical terms:

$$ax + bx = x(a + b)$$

Example:

$$3x^2 - 6x = 3x(x - 2)$$

2. Diff. of Squares

Recognize $A^2 - B^2$:

$$A^2 - B^2 = (A - B)(A + B)$$

Example:

$$x^2 - 9 = (x - 3)(x + 3)$$

3. Trinomials

Find numbers that multiply to c & add to b:

$$x^2 + bx + c = (x + p)(x + q)$$

Example:

$$u^2 - 5u + 6 = (u - 2)(u - 3)$$

Key Strategy

If x appears in multiple powers, always rearrange to make one side equal to 0, then factor!

Case 3: Factoring Techniques in Rearrangements

A) Grouping & Common Factor

Solve for x in $v^2 - x = u^2 + 2ax$:

1. Collect x-terms:

$$v^2 - u^2 = 2ax + x$$

2. Factor out x:

$$v^2 - u^2 = x(2a + 1)$$

3. Isolate x:

$$x = \frac{v^2 - u^2}{2a + 1}$$

B) Difference of Squares

Solve for x in $e^{2x} - 9 = 0$:

1. Recognize $A^2 - B^2 = (A - B)(A + B)$:

$$(e^x)^2 - 3^2 = 0$$

2. Factor into linear terms:

$$(e^x - 3)(e^x + 3) = 0$$

3. Solve each factor:

$$e^x = 3 \implies x = \ln(3)$$

$$e^x = -3 \implies \text{No real solution!}$$

Case 3: Quadratic Form via Substitution

C) Equations in Quadratic Form

Solve for x in $e^{2x} - 5e^x + 6 = 0$:

Step-by-Step Substitution

1. Identify $e^{2x} = (e^x)^2$ and set $u = e^x$:

$$u^2 - 5u + 6 = 0$$

2. Factor the trinomial:

$$(u-2)(u-3) = 0$$

3. Find roots for u:

$$u = 2 \quad \text{or} \quad u = 3$$

Back-Substitution & Final Solve

1. Replace u with e^x:

$$e^x = 2 \quad \text{and} \quad e^x = 3$$

2. Apply ln to isolate x:

$$x = \ln(2) \quad \text{and} \quad x = \ln(3)$$

Guided Practice (10 min)

Pair Challenge

Solve for x in each of the 3 cases using full algebraic steps in your notebook:

1. **Case 1 (Fractions):**

$$y = \frac{3x - 5}{2x + 4}$$

2. **Case 2 (Logarithm Properties):**

$$3^x = 7^{x-1}$$

3. **Case 3 (Factoring / Quadratic Form):**

$$e^{2x} - 4e^x = 0$$

Success Criteria

Show all intermediate steps. Explicitly point out where you factor or apply logarithm properties!

Session Synthesis: Which Strategy to Use?

Structure	Key Action	Outcome
Fraction $\frac{ax+b}{cx+d}$	Multiply by denominator	Collect x on one side & factor out x
Exponentials $a^x = b^{mx+c}$	Take ln on both sides & power law	Solve linear equation for x
Multiple powers $e^{2x}, x^2, \ldots$	Common factor or substitution $u = a^x$	Factor into linear terms

Great job mastering advanced rearrangements today!